6.876 Advanced Topics in Cryptography: Lattices	September 9, 2015

Lecture 1

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Lattices are amazing mathematical objects with applications all over mathematics and theoretical computer science. Examples include

- **Sphere Packing:** A classical problem called the "Sphere Packing Problem" asks for a way to pack the largest number of spheres of equal volume in 3-dimensional space (in an asymptotic sense, as the volume of the available space goes to infinity). The so-called *Kepler's Conjecture*, recently turned into a theorem by Hales, states that the face-centered cubic lattice offers the optimal packing of spheres in 3 dimensions.

  The optimal sphere packing in 2 and 3 dimensions are lattice packings – could this be the case in higher dimensions as well? This remains a mystery.

- **Error Correcting Codes:** Generalizing to n dimensions, the sphere packing problem and friends have applications to constructing *error-correcting codes* with the optimal rate.

- **Number Theory:** In mathematics, the study of lattices is called the "Geometry of Numbers", a term coined by Hermann Minkowski. Minkowski's Theorem and subsequent developments have had an enormous impact on Number Theory, Functional Analysis and Convex Geometry. Lattices have been used to test various number theoretic conjectures, the most famous being a disproof of Merten's Conjecture by Odlyzko and te Riele in 1985.

Lattices have also been quite influential in Theoretical Computer Science:

- In **Algorithms:** The famed Lenstra-Lenstra-Lovász algorithm for the shortest vector problem has generated a treasure-trove of algorithmic applications. Lattices have been used to construct an Integer Linear Programming algorithm in constant dimensions, in factoring polynomials over the rationals, and algorithms to find small solutions to systems of polynomial equations.

- In **Complexity Theory:** Lattices provide one of the most striking sources of problems with a worst-case to average-case connection. NP-hard problems are widely believed to be hard in the worst case, but are they hard on typical or average instances? For many problems and many average-case distributions, we know that this is not the case. In contrast, for the (approximate) shortest vector problem, we can show that finding a solution in a "random lattice" chosen from a certain easily sampleable distribution is as hard as finding a solution in the worst case, namely for arbitrary lattices.

- In **Cryptography:** The first applications of lattices in Cryptography have been in breaking cryptosystems, for example, variants of the knapsack cryptosystem, the NTRU cryptosystem and special cases of the RSA function. More recently, however, lattices have been used quite successfully in constructing secure cryptographic algorithms that achieve highly expressive functionalities such as fully homomorphic encryption.

In this course, we will study lattices from the point of view of theoretical computer science, first the mathematics of lattices, then the algorithms and complexity theory and finally lattice-based cryptography.

Notation. We will denote the natural numbers by $\mathbb{N}$, integers by $\mathbb{Z}$, rationals by $\mathbb{Q}$ and the reals by $\mathbb{R}$.

1 Lattices

Definition 1 (Lattices)**.** *Given n linearly independent vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^m$, the lattice generated by them is defined as*

$$\mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n) \stackrel{\text{def}}{=} \left\{ \sum_{i=1}^{n} x_i \mathbf{b}_i \mid x_i \in \mathbb{Z} \right\}$$

We call $\mathbf{b}_1, \ldots, \mathbf{b}_n$ a *basis* of the lattice. Note that the definition requires $\mathbf{b}_1, \ldots, \mathbf{b}_n$ to be linearly independent over $\mathbb{R}$ (and not over $\mathbb{Z}$).

We call n the *rank* of the lattice, and m the *dimension* of the lattice. In general, $n \leq m$. When $n = m$, we call the lattice a *full-rank* lattice. Throughout this course, we will focus on full-rank lattices – most results we prove can be generalized to the non full-rank case.

We will use a notational short-hand when dealing with bases, denoting them by a matrix $\mathbf{B}$ whose columns are the basis vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$. That is, we will write

$$\mathbf{B} = \begin{pmatrix} | & & | \\ \mathbf{b}_1 & \ldots & \mathbf{b}_n \\ | & & | \end{pmatrix}$$

and thus, in this notation,

$$\mathcal{L}(\mathbf{B}) \stackrel{\text{def}}{=} \{\mathbf{B}\mathbf{x} \mid \mathbf{x} \in \mathbb{Z}^n\}$$

In general, we treat all vectors as column vectors unless otherwise specified. For a matrix $\mathbf{B}$ (resp. row vector $\mathbf{v}$), $\mathbf{B}^T$ (resp. $\mathbf{v}^T$) denotes the transpose of $\mathbf{B}$ (resp. $\mathbf{v}$).

Examples of Lattices.

1. Figure 1(a) shows the lattice in 2 dimensions generated by the vectors $(1,0)^T$ and $(0,1)^T$. This lattice is the set of all points in $\mathbb{R}^2$ with integer coordinates.

   This can be generalized to n dimensions, where the lattice $\mathbb{Z}^n$ is called the *integer lattice*.

2. Figure 1(b) shows a different basis for the same lattice, namely the basis consisting of the vectors $(1,2)^T$ and $(2,3)^T$.

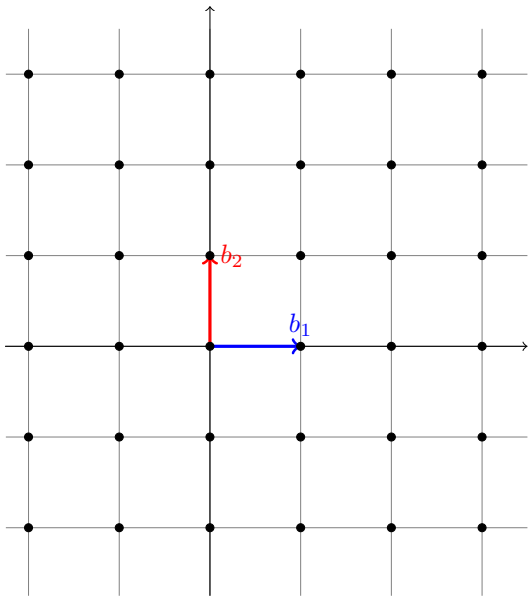
3. Figure 1(c) shows a different lattice in 2 dimensions, generated by the basis vectors $(2,0)^T$ and $(1,1)^T$. Note that this is a sub-lattice of $\mathbb{Z}^2$, namely a subset of $\mathbb{Z}^2$ which is also a lattice. (We will formally define sublattices later in the course).

4. In one dimension, all lattices are multiples of a single number. For example, the lattice generated by (2) is the set of all even numbers.

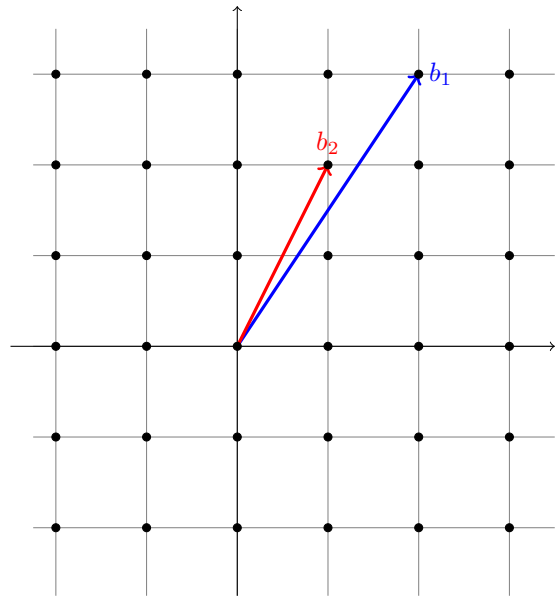
5. All the examples we saw so far are full-rank lattices. Figure 1(d) shows a lattice in 2 dimensions generated by the vector $(1,1)^T$ – this lattice has rank 1. We will not deal with non full-rank lattices in this course.

6. The set of points generated by (1) and $(\sqrt{2})$ in one dimension is not a lattice. First, this example does not conform to Definition 1 since 1 and $\sqrt{2}$ are *linearly dependent* over $\mathbb{R}$. Secondly, any n-dimensional lattice is a discrete subset of $\mathbb{Z}^n$ (see Lecture 2 for why this is the case). However, the set generated by (1) and $(\sqrt{2})$ is not a discrete subset of $\mathbb{Z}$ since one can generate arbitrarily small numbers as linear combinations of 1 and $\sqrt{2}$.

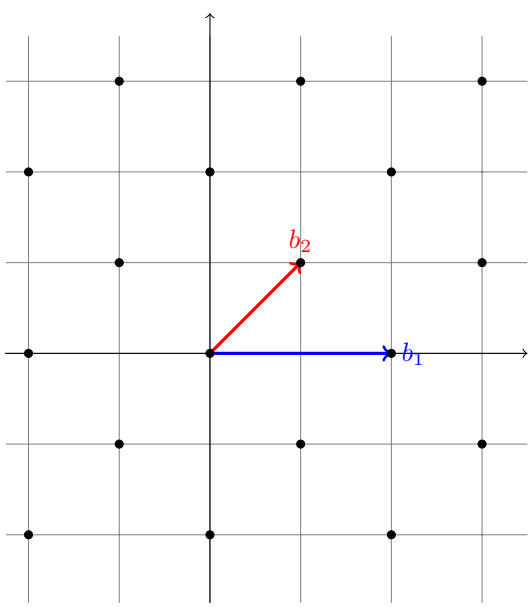
It is instructive to compare the definition of a lattice generated by n linearly independent vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$ to the definition of the span of these vectors.



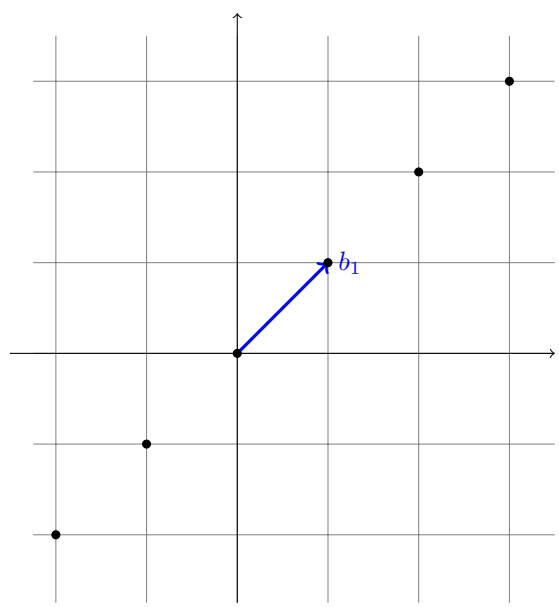
(a) The lattice $\mathbb{Z}^2$ with basis vectors $(0, 1)$ and $(1, 0)$.



(b) The lattice $\mathbb{Z}^2$ with a different basis consisting of vectors $(1, 2)$ and $(2, 3)$. In fact, any lattice has infinitely many bases.



(c) A full-rank lattice generated by the basis vectors $(1, 1)$ and $(2, 0)$. Note that this is a sub-lattice of $\mathbb{Z}^2$.



(d) A *non full-rank* lattice with basis vector $(1, 1)$

Figure 1: Various lattices and their bases.

Definition 2 (Span)**.** *Given n* linearly independent *vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^m$, their* span *is defined as*

$$\mathsf{Span}(\mathbf{b}_1, \ldots, \mathbf{b}_n) \stackrel{\text{def}}{=} \left\{ \sum_{i=1}^{n} x_i \mathbf{b}_i \mid x_i \in \mathbb{R} \right\}$$

Note the difference between Definition 1 of a lattice generated by a set of vectors – which consists of all of its *integer* linear combinations – and the above definition of the span of a set of vectors – which consists of all of its linear combinations with *real* coefficients. The crucial power of lattices comes from the fact that it is a discrete set (which the span is not).

Clearly, $\mathsf{Span}(\mathbf{b}_1, \ldots, \mathbf{b}_n) \supset \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$.

2 Same Lattice, Many Bases

We already saw from the examples above (Figure 1(a) and Figure 1(b)) that the same lattice can have many different bases. For example, it turns out that all the bases given below generate the same lattice, namely $\mathbb{Z}^2$:

$$\mathbf{B}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{B}_2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{B}_3 = \begin{pmatrix} 647 & 64 \\ 91 & 9 \end{pmatrix}$$

but the following basis *does not* generate $\mathbb{Z}^2$, but only a proper *sub-lattice* of $\mathbb{Z}^2$.

$$\mathbf{B}_4 = \begin{pmatrix} 42 & 41 \\ 9 & 8 \end{pmatrix}$$

In fact, any lattice has infinitely many bases. In particular, the bases can have arbitrarily large coefficients.

A natural question to ask is: *how can we efficiently tell if two given bases $\mathbf{B}$ and $\mathbf{B}'$ generate the same lattice?* We will give two answers to this question – an algebraic answer and a geometric answer.

2.1 An Algebraic Characterization using Unimodular Matrices

Our first characterization provides an efficient algorithm to determine if two bases generate the same lattice. In order to present the characterization, we first need to define the notion of a *unimodular matrix*.

Notation. For any $x \in \mathbb{R}$, we will let $|x|$ represent the absolute value of x.

Definition 3. *A matrix $\mathbf{U} \in \mathbb{Z}^{n \times n}$ is unimodular if $|\det(\mathbf{U})| = 1$.*

Here, $\det(\mathbf{U})$ denotes the determinant of the (square) matrix $\mathbf{U}$, and $|\cdot|$ denotes the absolute value. For example, the matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ is unimodular, and so is $\left(\begin{smallmatrix} 647 & 64 \\ 91 & 9 \end{smallmatrix}\right)$, but not $\left(\begin{smallmatrix} 42 & 41 \\ 9 & 8 \end{smallmatrix}\right)$.

Proposition 4. *If $\mathbf{U}$ is unimodular, so is $\mathbf{U}^{-1}$.*

Proof. This follows from the way inverses are computed. In particular,

- Each entry in $\mathbf{U}^{-1}$ is the ratio of the determinant of a *minor* of $\mathbf{U}$ to the determinant of $\mathbf{U}$ itself. Since the determinant of any minor of $\mathbf{U}$ is an integer, and the determinant of $\mathbf{U}$ is ± 1, each entry of $\mathbf{U}^{-1}$ is an integer. Thus, $\mathbf{U}^{-1} \in \mathbb{Z}^{n \times n}$.
- $\det(\mathbf{U}^{-1}) = 1/\det(\mathbf{U}) = \pm 1$. Thus, $|\det(\mathbf{U}^{-1})| = 1$.

Together, these two observations mean that $\mathbf{U}$ is unimodular. □

We can now state the characterization of equivalent bases.

Theorem 5. *Given two full-rank bases $\mathbf{B} \in \mathbb{R}^{n \times n}$ and $\mathbf{B}' \in \mathbb{R}^{n \times n}$, the following two conditions are equivalent:*

- $\mathcal{L}(\mathbf{B}) = \mathcal{L}(\mathbf{B}')$
- *There exists a unimodular matrix* $\mathbf{U}$ *such that* $\mathbf{B}' = \mathbf{B}\mathbf{U}$.

Proof. ("$\Rightarrow$") First, assume that $\mathcal{L}(\mathbf{B}) = \mathcal{L}(\mathbf{B}')$. Then, there are integer matrices $\mathbf{V}$ and $\mathbf{V}'$ such that

$$\mathbf{B}' = \mathbf{B}\mathbf{V} \qquad \text{and} \qquad \mathbf{B} = \mathbf{B}'\mathbf{V}'$$

It suffices to show that $|\det(\mathbf{V})| = |\det(\mathbf{V}')| = 1$.

Putting these two equations together, we have $\mathbf{B}' = \mathbf{B}\mathbf{V} = \mathbf{B}'(\mathbf{V}'\mathbf{V})$. Since $\mathbf{B}'$ is non-singular (remember: $\mathbf{B}$ is a full-rank matrix, and so is $\mathbf{B}'$) we can multiply both sides of the equation by $(\mathbf{B}')^{-1}$ and we get

$$\mathbf{V}'\mathbf{V} = \mathbf{1}_n \tag{1}$$

were $\mathbf{1}_n$ denotes the n-by-n identity matrix.

Since determinant is multiplicative, we get $\det(\mathbf{V}')\det(\mathbf{V}) = 1$. Since $\mathbf{V}$ and $\mathbf{V}'$ are integer matrices, their determinant is also an integer.

Putting these two facts together, we see that the only two choices are:

- $\det(\mathbf{V}) = \det(\mathbf{V}') = 1$, or
- $\det(\mathbf{V}) = \det(\mathbf{V}') = -1$

In either case, $|\det(\mathbf{V})| = |\det(\mathbf{V}')| = 1$, and we are done.

("$\Leftarrow$") For the other direction, assume that there is a unimodular matrix $\mathbf{U}$ such that $\mathbf{B}' = \mathbf{B}\mathbf{U}$. Then, since $\mathbf{U}$ is an integer matrix,

$$\mathcal{L}(\mathbf{B}') \subseteq \mathcal{L}(\mathbf{B})$$

This is because each vector (column) of $\mathbf{B}'$ can be written as a linear combination of vectors in $\mathbf{B}$. Thus, the set of all integer linear combinations of vectors in $\mathbf{B}'$ is contained in the set of all integer linear combinations of vectors in $\mathbf{B}$.

Now, $\mathbf{B} = \mathbf{B}'(\mathbf{U}^{-1})$ where $\mathbf{U}^{-1}$ is also unimodular by Proposition 4. This shows that

$$\mathcal{L}(\mathbf{B}) \subseteq \mathcal{L}(\mathbf{B}')$$

by the same argument as above. Together, we have $\mathcal{L}(\mathbf{B}) = \mathcal{L}(\mathbf{B}')$. □

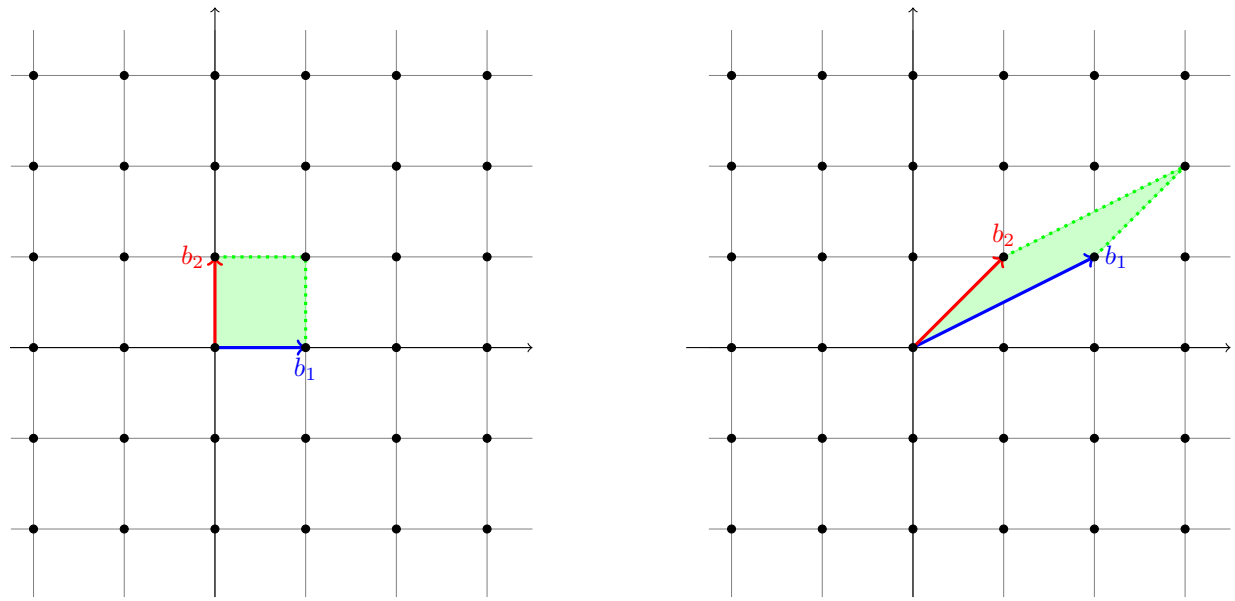
2.2 A Geometric Characterization using the Fundamental Parallelepiped

We need the notion of a fundamental parallelepiped of a basis $\mathbf{b}_1, \ldots, \mathbf{b}_n$.

Definition 6 (Fundamental Parallelepiped)**.** *Given* n linearly independent *vectors* $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^m$*, their* fundamental parallelepiped *is defined as*

$$\mathcal{P}(\mathbf{b}_1, \ldots, \mathbf{b}_n) \stackrel{\text{def}}{=} \left\{ \sum_{i=1}^{n} x_i \mathbf{b}_i \mid x_i \in \mathbb{R}, 0 \leq x_i < 1 \right\}$$

Thus, pictorially, a fundamental parallelepiped is the (half-open) region enclosed by the vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$. Clearly, different bases of the same lattice generate different fundamental paralellepipeds. See Figure 2(a) and 2(b).



(a) The lattice $\mathbb{Z}^2$ with basis vectors $(0, 1)$ and $(1, 0)$ and the associated fundamental parallelepiped.

(b) The lattice $\mathbb{Z}^2$ with a different basis consisting of vectors $(1, 1)$ and $(2, 1)$, and the associated fundamental parallelepiped.

Figure 2: Parallelepipeds for various bases of the lattice $\mathbb{Z}^2$. Note that the parallelepipeds in either case do not contain any non-zero lattice point.

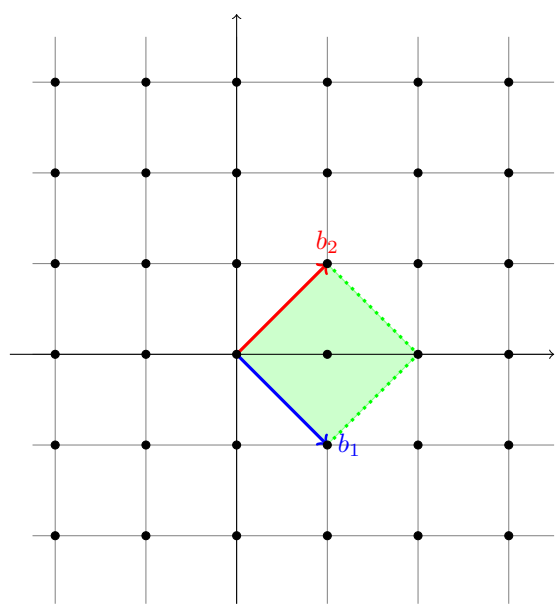


Figure 3: $\mathbf{b}_1$ and $\mathbf{b}_2$ do not form a basis of $\mathbb{Z}^2$. Note that the parallelepiped of $\mathbf{b}_1$ and $\mathbf{b}_2$ contains a non-zero lattice point, namely $(1, 0)$.

Note that in Figures 2(a) and 2(b), the vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ form a basis of the lattice, and the parallelepiped associated to the basis does not contain any lattice point other than $\mathbf{0}$. On the other hand, in Figure 3, the vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ do not form a basis of the lattice, and the parallelepiped associated to the basis contains a non-zero lattice point. In fact, this is not a coincidence as our next theorem shows.

Theorem 7. *Let $\mathcal{L}$ be a full-rank n-dimensional lattice, and let $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^n$ denote* linearly independent *vectors in $\mathcal{L}$. Then, $\mathbf{b}_1, \ldots, \mathbf{b}_n$ form a basis of $\mathcal{L}$* if and only if $\mathcal{P}(\mathbf{b}_1, \ldots, \mathbf{b}_n) \cap \mathcal{L} = \{\mathbf{0}\}$.

Proof. ("$\Rightarrow$") Suppose that $\mathbf{b}_1, \ldots, \mathbf{b}_n$ is a basis of $\mathcal{L}$. Let

$$\mathbf{a} = \sum_{i=1}^{n} x_i \mathbf{b}_i \in \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n) \cap \mathcal{P}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$$

We will show that $\mathbf{a} = \mathbf{0}$.

Since $\mathbf{a} \in \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$, $x_i \in \mathbb{Z}$ for all i. Since $\mathbf{a} \in \mathcal{P}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$, $x_i \in [0, 1)$ for all i. Together, this means that $x_i = 0$ for all i, and thus, $\mathbf{a} = \mathbf{0}$.

("$\Leftarrow$") Suppose that $\mathcal{P}(\mathbf{b}_1, \ldots, \mathbf{b}_n) \cap \mathcal{L} = \{\mathbf{0}\}$. We would like to show that $\mathbf{b}_1, \ldots, \mathbf{b}_n$ form a basis of $\mathcal{L}$.

The vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$ are linearly independent. Since they belong to $\mathcal{L}$, $\mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n) \subseteq \mathcal{L}$. What remains is to show that $\mathcal{L} \subseteq \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$. Pick any vector $\mathbf{a} \in \mathcal{L}$ and write it as

$$\mathbf{a} = \sum_{i=1}^{n} x_i \mathbf{b}_i \qquad \text{where } x_i \in \mathbb{R}$$

Consider now the vector

$$\mathbf{a}' = \sum_{i=1}^{n} \lfloor x_i \rfloor \, \mathbf{b}_i \qquad \in \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$$

which is clearly in the lattice $\mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$ since the coefficients $\lfloor x_i \rfloor$ are integers. Therefore, the vector $\mathbf{a} - \mathbf{a}'$ is in $\mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$ as well. Now,

$$\mathbf{a} - \mathbf{a}' = \sum_{i=1}^{n} (x_i - \lfloor x_i \rfloor) \mathbf{b}_i \qquad \in \mathcal{P}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$$

is in the parallelepiped of $\mathbf{b}_1, \ldots, \mathbf{b}_n$ since $0 \leq x_i - \lfloor x_i \rfloor < 1$ for all i.

Since $\mathbf{a} - \mathbf{a}' \in \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n) \cap \mathcal{P}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$, it must be the case that $\mathbf{a} - \mathbf{a}' = 0$ by assumption. Since the vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$ are linearly independent, this means that $x_i - \lfloor x_i \rfloor = 0$ for all i which in turn means that $x_i \in \mathbb{Z}$ for all i.

Thus, $\mathbf{a} \in \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$, showing us that $\mathcal{L} \subseteq \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_n)$. □

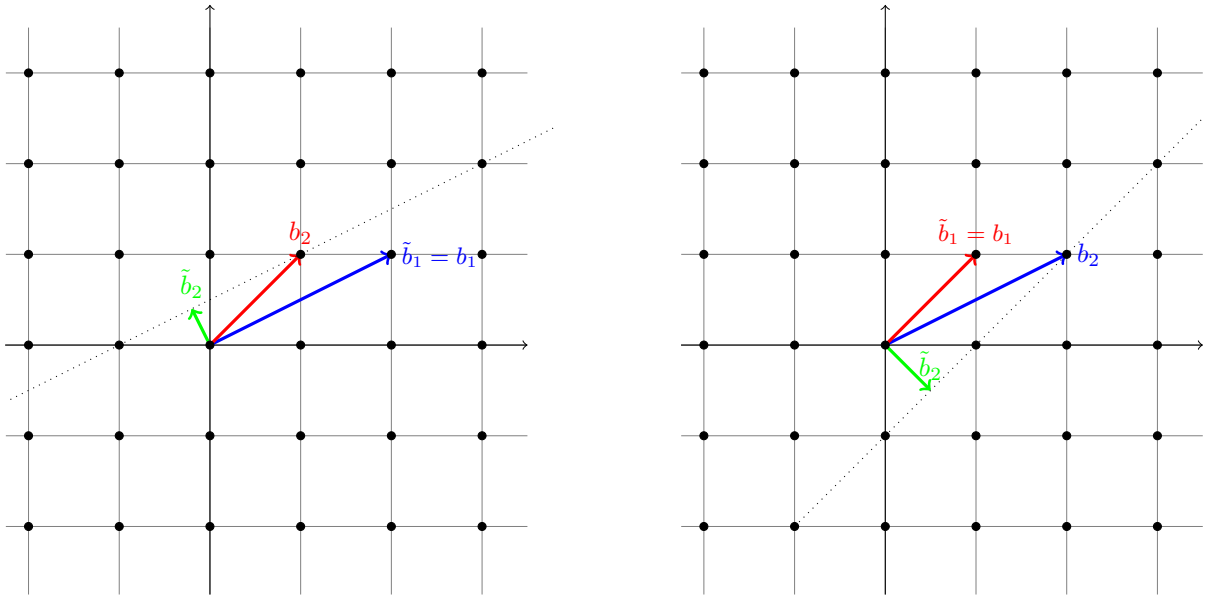
2.3 Determinant of a Lattice

Another quantity associated to a lattice is its determinant, denoted $\det(\mathcal{L})$. The determinant of a lattice is the n-dimensional volume of its fundamental parallelepiped, computed as the absolute value of the determinant of its basis matrix $\mathbf{B}$. A couple of facts about the determinant of a lattice are worth noting:

1. The parallelepipeds associated with different bases of a lattice have the same volume. Thus, *the determinant is a lattice invariant.* This is easy to see using our characterization of equivalent bases from Theorem 5.

   Let $\mathbf{B}$ and $\mathbf{B}'$ be any two lattice bases. By Theorem 5, there is a unimodular matrix $\mathbf{U}$ such that $\mathbf{B}' = \mathbf{B}\mathbf{U}$. Thus, $|\det(\mathbf{B}')| = |\det(\mathbf{B})| \cdot |\det(\mathbf{U})| = |\det(\mathbf{B})|$ since $|\det(\mathbf{U})| = 1$.

2. Intuitively, the determinant of a lattice is inversely proportional to its "density". The larger the determinant, the sparser the lattice.



(a) Gram-Schmidt orthogonalization of the vectors b_1 and b_2 in that order.

(b) Gram-Schmidt orthogonalization of the same vectors, but in the opposite order.

Figure 4: Gram-Schmidt Orthogonalization.

3 Gram-Schmidt Orthogonalization

Gram-Schmidt orthogonalization is a procedure in linear algebra that transforms a set of vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$ into a set of orthogonal vectors $\tilde{\mathbf{b}}_1, \ldots, \tilde{\mathbf{b}}_n$. In two dimensions, this proceeds as follows:

- The first Gram-Schmidt vector $\tilde{\mathbf{b}}_1$ is $\mathbf{b}_1$ itself.
- The second Gram-Schmidt vector $\tilde{\mathbf{b}}_2$ is the component of $\mathbf{b}_2$ that is orthogonal to $\mathsf{Span}(\tilde{\mathbf{b}}_1)$. This can be computed as

  $$\tilde{\mathbf{b}}_2 = \mathbf{b}_2 - \left(\frac{\langle \mathbf{b}_2, \tilde{\mathbf{b}}_1 \rangle}{\langle \tilde{\mathbf{b}}_1, \tilde{\mathbf{b}}_1 \rangle}\right)\tilde{\mathbf{b}}_1$$

  See Figure 4 for an illustration of this process.

In general, the Gram-Schmidt vectors are obtained by projecting each vector successively on the space orthogonal to the span of all the previous vectors.

Definition 8 (Gram-Schmidt Orthogonalization)**.** *For a sequence of n linearly independent vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^n$, we define their* Gram-Schmidt orthogonalization *as the sequence of vectors $\tilde{\mathbf{b}}_1, \ldots, \tilde{\mathbf{b}}_n$ defined as follows:*

$$\tilde{\mathbf{b}}_i = \mathbf{b}_i - \sum_{j=1}^{i-1} \mu_{i,j} \tilde{\mathbf{b}}_j \qquad \textit{where } \mu_{i,j} = \frac{\langle \mathbf{b}_i, \tilde{\mathbf{b}}_j \rangle}{\langle \tilde{\mathbf{b}}_j, \tilde{\mathbf{b}}_j \rangle}$$

Thus, $\tilde{\mathbf{b}}_j$ is the component of $\mathbf{b}_i$ that is orthogonal to $\tilde{\mathbf{b}}_1, \ldots, \tilde{\mathbf{b}}_{i-1}$. The coefficients $\mu_{i,j}$ are called the Gram-Schmidt coefficients.

Remarks.

1. True to its name, the different Gram-Schmidt vectors $\tilde{\mathbf{b}}_1, \ldots, \tilde{\mathbf{b}}_n$ are orthogonal to each other. That is, for each $i \neq j$, $\langle \tilde{\mathbf{b}}_i, \tilde{\mathbf{b}}_j \rangle = 0$. This is an easy consequence of Definition 8.

2. The span of $\tilde{\mathbf{b}}_1, \ldots, \tilde{\mathbf{b}}_i$ is the same as the span of $\mathbf{b}_1, \ldots, \mathbf{b}_i$ for all $1 \leq i \leq n$.

3. The vectors $\tilde{\mathbf{b}}_1, \ldots, \tilde{\mathbf{b}}_n$ do not form a lattice basis. In fact, the Gram-Schmidt vectors are not necessarily in the lattice. See Figure 4 for example.

4. The (Euclidean) length of the Gram-Schmidt vector $\tilde{\mathbf{b}}_i$ is at most the length of the basis vector $\mathbf{b}_i$. Namely, $\|\tilde{\mathbf{b}}_i\| \leq \|\mathbf{b}_i\|$.

5. Clearly, as seen in Figure 4, the Gram-Schmidt vectors depend on the order in which the vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$ are processed.

Let $\tilde{\mathbf{b}}_1/\|\tilde{\mathbf{b}}_1\|, \ldots, \tilde{\mathbf{b}}_n/\|\tilde{\mathbf{b}}_n\|$ denote the unit vectors in the direction of the Gram-Schmidt vectors. Then, the Gram-Schmidt orthogonalization process can be written in matrix form as

$$\begin{aligned}
\begin{pmatrix} | & & | \\ \mathbf{b}_1 & \ldots & \mathbf{b}_n \\ | & & | \end{pmatrix} &= \begin{pmatrix} | & & | \\ \tilde{\mathbf{b}}_1 & \ldots & \tilde{\mathbf{b}}_n \\ | & & | \end{pmatrix} \cdot \begin{pmatrix} 1 & \mu_{2,1} & \mu_{3,1} & \ldots & \mu_{n,1} \\ 0 & 1 & \mu_{3,2} & \ldots & \mu_{n,2} \\ 0 & 0 & 1 & \ldots & \mu_{n,3} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} | & & | \\ \frac{\tilde{\mathbf{b}}_1}{\|\tilde{\mathbf{b}}_1\|} & \ldots & \frac{\tilde{\mathbf{b}}_n}{\|\tilde{\mathbf{b}}_n\|} \\ | & & | \end{pmatrix} \cdot \begin{pmatrix} \|\tilde{\mathbf{b}}_1\| & \mu_{2,1}\|\tilde{\mathbf{b}}_1\| & \mu_{3,1}\|\tilde{\mathbf{b}}_1\| & \ldots & \mu_{n,1}\|\tilde{\mathbf{b}}_1\| \\ 0 & \|\tilde{\mathbf{b}}_2\| & \mu_{3,2}\|\tilde{\mathbf{b}}_2\| & \ldots & \mu_{n,2}\|\tilde{\mathbf{b}}_2\| \\ 0 & 0 & \|\tilde{\mathbf{b}}_3\| & \ldots & \mu_{n,3}\|\tilde{\mathbf{b}}_3\| \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \|\tilde{\mathbf{b}}_n\| \end{pmatrix}
\end{aligned}$$

Since the vectors $\frac{\tilde{\mathbf{b}}_i}{\|\tilde{\mathbf{b}}_i\|}$ are orthonormal, the determinant of the matrix with columns $\frac{\tilde{\mathbf{b}}_i}{\|\tilde{\mathbf{b}}_i\|}$ is 1.

Thus, we have

$$\det(\mathcal{L}(\mathbf{B})) = \prod_{i=1}^{n} \|\tilde{\mathbf{b}}_i\|$$

In other words, the Gram-Schmidt orthogonalization process is a volume-preserving transformation that results in a set of orthogonal vectors $\tilde{\mathbf{b}}_1, \ldots, \tilde{\mathbf{b}}_n$, whose enclosing parallelepiped is rectangular and generates a volume of $\prod_{i=1}^{n} \|\tilde{\mathbf{b}}_i\|$.

4 Successive Minima of a Lattice

A basic parameter of the lattice is the length of the shortest non-zero vector in the lattice (since any lattice contains the zero vector which has norm zero, we have to ask for a non-zero vector). This parameter is also called the *first successive minimum* of the lattice, and is denoted $\lambda_1(\mathcal{L})$. When we speak of length, we mean the Euclidean norm defined as follows: for a vector $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, the Euclidean norm of $\mathbf{x}$, denoted $\|\mathbf{x}\|_2$ (or simply as $\|\mathbf{x}\|$ is defined as

$$\|\mathbf{x}\| = \sqrt{\sum_{i=1}^{n} x_i^2}$$

The Euclidean norm is also frequently referred to as the ℓ_2 norm. We can speak of other norms such as the ℓ_1 norm – $\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i|$ – and the ℓ_∞ norm – $\|\mathbf{x}\|_\infty = \max_{i=1}^{n} |x_i|$, but we will stick to the Euclidean norm for most of this course.

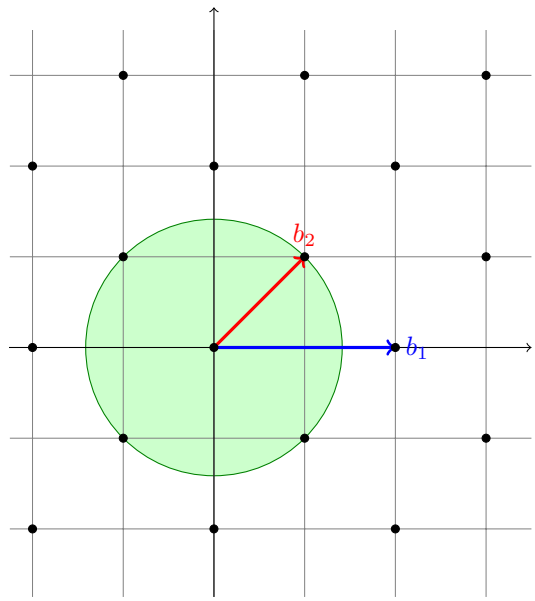


Figure 5: The shortest vector in the lattice generated by $(1, 1)$ and $(2, 0)$. $\lambda_1(\mathcal{L}) = \sqrt{2}$.

Figure 5 shows a shortest vector in the lattice generated by $(1, 1)$ and $(2, 0)$. The shortest vector is not unique in general. There could be many, even exponentially many, shortest vectors. Clearly, there are at least two – if $\mathbf{v}$ is a shortest vector in a lattice, then so is $-\mathbf{v}$.

We will be interested in lower and upper bounds on λ_1. We first show a lower bound on λ_1 using Gram-Schmidt orthogonalization. In the next lecture, we will prove Minkowski's theorem which provides an upper bound on λ_1 in terms of the determinant of the lattice.

Lower Bound on λ_1. We show the following theorem. Roughly speaking, the theorem says that the shortest non-zero vector in a lattice is at least as long as the shortest Gram-Schmidt vector of a basis of the lattice. To see why, observe that a lattice can be partitioned into many hyperplanes perpendicular to its Gram-Schmidt vector $\tilde{\mathbf{b}}_n$. See Figure 6 for an illustration in two dimensions.

Now, there are two possibilities:

- There is a shortest non-zero vector in one of the hyper-planes not passing through the origin. In that case, the vector has to have length at least $\|\tilde{\mathbf{b}}_n\| \geq \min_j \|\tilde{\mathbf{b}}_j\|$ since the ith such hyper-plane is at a distance of $i \cdot \|\tilde{\mathbf{b}}_n\|$ from the origin.
- The shortest non-zero vector lives in the hyper-plane that passes through the origin, in which case, repeat the same argument in dimension $n-1$ with the $(n-1)$-dimensional sublattice partitioned into hyper-planes perpendicular to $\tilde{\mathbf{b}}_{n-1}$.

Eventually, if the argument reaches dimension 1, the shortest non-zero vector has to have length at least $\|\mathbf{b}_1\| = \|\tilde{\mathbf{b}}_1\| \geq \min_j \|\tilde{\mathbf{b}}_j\|$.

The formal statement and proof of the theorem follows.

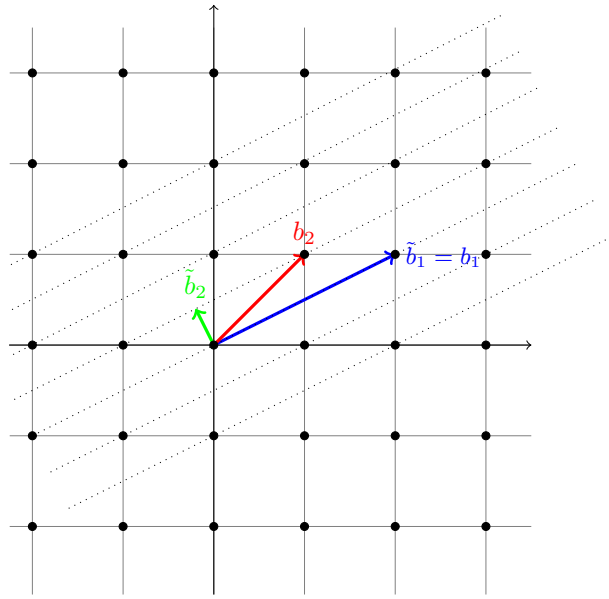


Figure 6: The lattice is partitioned into many parallel hyperplanes perpendicular to $\tilde{\mathbf{b}}_2$. Either the shortest vector lives in a hyperplane that does not pass through the origin, in which case its length is at least $\|\tilde{\mathbf{b}}_2\|$ or it lives in the hyperplane that passes through the origin, in which case its length is at least $\tilde{\mathbf{b}}_1 = \|\tilde{\mathbf{b}}_1\|$. In general, in two dimensions, $\lambda_1(\mathcal{L}) \geq \min\{\|\tilde{\mathbf{b}}_1\|, \|\tilde{\mathbf{b}}_2\|\}$. This argument can be generalized to n dimensions.

Theorem 9. *Let* $\mathbf{B}$ *be a rank-n lattice basis, and* $\tilde{\mathbf{B}}$ *be its Gram-Schmidt orthogonalization. Then,*

$$\lambda_1(\mathcal{L}(\mathbf{B})) \geq \min_{i=1,\ldots,n} \|\tilde{\mathbf{b}}_i\| > 0$$

Proof. Let $\mathbf{x} \in \mathbb{Z}^n$ be *any* non-zero integer vector. We would like to show that the lattice vector $\mathbf{Bx} \in \mathcal{L}(\mathbf{B})$ has length at least $\min_i \|\tilde{\mathbf{b}}_i\|$.

The proof follows by calculating the quantity $|\langle \mathbf{Bx}, \tilde{\mathbf{b}}_j \rangle|$ in two different ways.

1. Let $j \in \{1, \ldots, n\}$ be the largest index such that $x_j \neq 0$. Then,

$$|\langle \mathbf{Bx}, \tilde{\mathbf{b}}_j \rangle| = |\langle \sum_{i=1}^{n} x_i \mathbf{b}_i, \tilde{\mathbf{b}}_j \rangle| = |\sum_{i=1}^{n} x_i \langle \mathbf{b}_i, \tilde{\mathbf{b}}_j \rangle| = |x_j| \langle \tilde{\mathbf{b}}_j, \tilde{\mathbf{b}}_j \rangle = |x_j| \cdot \|\tilde{\mathbf{b}}_j\|^2 \tag{2}$$

where the first equality follows by rewriting $\mathbf{Bx}$ as $\sum_{i=1}^{n} x_i \mathbf{b}_i$, the second follows by the linearity of the inner product, and the third because

- for $j < i$, $\langle \mathbf{b}_i, \tilde{\mathbf{b}}_j \rangle = 0$
- for $j > i$, $x_j = 0$ by the definition of j.

The fourth equality follows by the definition of $\|\tilde{\mathbf{b}}_j\|^2 = \langle \tilde{\mathbf{b}}_j, \tilde{\mathbf{b}}_j \rangle$.

2. On the other hand,

$$|\langle \mathbf{Bx}, \tilde{\mathbf{b}}_j \rangle| \leq \|\mathbf{Bx}\| \cdot \|\tilde{\mathbf{b}}_j\| \tag{3}$$

by the Cauchy-Schwarz inequality.

Putting together Equations 2 and 3, we get

$$\|\mathbf{Bx}\| \geq \frac{|\langle \mathbf{Bx}, \tilde{\mathbf{b}}_j \rangle|}{\|\tilde{\mathbf{b}}_j\|} = |x_j| \cdot \|\tilde{\mathbf{b}}_j\| \geq \|\tilde{\mathbf{b}}_j\| \geq \min_{i=1\ldots n} \|\tilde{\mathbf{b}}_i\|$$

where the third inequality follows from the fact that x_j is a non-zero integer. Since the length of any lattice vector is at least $\min_i \|\tilde{\mathbf{b}}_i\|$,

$$\lambda_1(\mathbf{B}) \geq \min_{i=1\ldots n} \|\tilde{\mathbf{b}}_i\|$$

Since $\mathbf{b}_1, \ldots, \mathbf{b}_n$ are linearly independent, this quantity is strictly positive. □

A corollary of this theorem is that a lattice is a *discrete set*. In other words, lattice points cannot be arbitrarily close to one another. Formally:

Corollary 10. *For every lattice $\mathcal{L}$, there is an $\epsilon = \epsilon(\mathcal{L}) > 0$ such that $\|\mathbf{x} - \mathbf{y}\| \geq \epsilon$ for any two unequal lattice points $\mathbf{x}, \mathbf{y} \in \mathcal{L}$.*

Proof. For any two $\mathbf{x} \neq \mathbf{y} \in \mathcal{L}$, $\mathbf{x} - \mathbf{y} \in \mathcal{L}$. Then, $\|\mathbf{x} - \mathbf{y}\| \geq \lambda_1(\mathcal{L}) > 0$. In particular, set $\epsilon = \lambda_1(\mathcal{L})$ to obtain the statement of the corollary. □

In fact, this leads us to a *basis-independent* characterization of a lattice. Namely, every discrete subset of $\mathbb{R}^n$ that is closed under subtraction is a lattice.

6.876 Advanced Topics in Cryptography: Lattices	September 14, 2015

Lecture 2

Lecturer: Vinod Vaikuntanathan *Scribe: Sergey Gorbunov*

1 Lecture Outline

- Alternative Definition of Lattices
- Upper bound on the length of a shortest lattice vector ()
- Successive Minima
- Applications of Minkowski's Theorem

2 Alternative Definition of Lattices

Last lecture we saw the following definition of a lattice:

Definition 1 (Lattices)**.** *Given n* linearly independent *vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^m$, the lattice generated by them is defined as*

$$(\mathbf{b}_1, \ldots, \mathbf{b}_n) \stackrel{\mathsf{def}}{=} \left\{ \sum_{i=1}^{n} x_i \mathbf{b}_i \mid x_i \in \mathbb{Z} \right\}$$

Now, consider the following set S:

$$S = \{(\mathbf{x}_1, \ldots, \mathbf{x}_n) \in : \sum_{i=1}^{n} \mathbf{x}_i \text{ is even}\}$$

A typical question may ask whether S is a lattice. From the Definition 1, we know that if we are able to find a set of linearly independent vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n \in$ such that for all $\mathbf{s} \in S$, $\mathbf{s} = a_1\mathbf{b}_1 + \ldots + a_n\mathbf{b}_n$ for some integer coefficients $a_1, \ldots, a_n$, then S is indeed a lattice. We will now give a new definition of a lattice that gives us a simpler way of verifying that S is a lattice.

Definition 2. *A lattice is a discrete additive subgroup of .*

In the above definition, by discrete we mean that:

$$\exists \epsilon > 0, \text{ s.t. } \forall \mathbf{x} \neq \mathbf{y} \in, ||\mathbf{x} - \mathbf{y}|| \geq \epsilon.$$

And by additive subgroup we mean that:

$$\forall \mathbf{x}, \mathbf{y} \in, \mathbf{x} - \mathbf{y} \in .$$

Claim 3. *Definition 1 is equivalent to Definition 2.*

Proof. We will first show that Definition 1 $\implies$ Definition 2.

Assume is a lattice defined as the set of all integer combinations of vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n \in$ which are linearly independent (Definition 1). Then, clearly is an additive subgroup of . In addition, $\forall \mathbf{x}, \mathbf{y} \in$, $\mathbf{x} - \mathbf{y} \in$. Therefore, from the lower bound on a shortest lattice vector,

$$||\mathbf{x} - \mathbf{y}|| \geq \lambda_1() \geq \min_{i=1,\ldots,n} ||\tilde{\mathbf{b}}_i||.$$

In other words, the length of any lattice vector must be greater than the length of a shortest lattice vector. Therefore, we can let $\epsilon = \lambda_1$. So, both properties of Definition 2 are satisfied (is a discrete additive subgroup of).

We show that Definition 2 $\implies$ Definition 1. Given a discrete additive subgroup of , we construct a set of basis using the algorithm below.

We will use the following definition of a closed parallelepiped.

Definition 4 (Closed Fundamental Parallelepiped)**.** *Given n* linearly independent *vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^m$, their* closed fundamental parallelepiped *is defined as*

$$(\mathbf{b}_1, \ldots, \mathbf{b}_n) \stackrel{\text{def}}{=} \left\{ \sum_{i=1}^{n} x_i \mathbf{b}_i \mid x_i \in \mathbb{R}, 0 \leq x_i \leq 1 \right\}$$

Pick $\mathbf{y} \in$ such that there is no lattice vector between the zero vector and $\mathbf{y}$. Let $\mathbf{b}_1 = \mathbf{y}$.
Iterate for all i, $1 \leq i < n$: Assume we have already chosen $\mathbf{b}_1, \ldots, \mathbf{b}_i$. Choose $\mathbf{y}$ not in the $(\mathbf{b}_1, \ldots, \mathbf{b}_i)$.
Consider a $(\mathbf{b}_1, \ldots, \mathbf{b}_i, \mathbf{y})$ (See Figure-1 for an example).
Now, contains at least one lattice point (namely $\mathbf{y}$) and it contains finitely many lattice points. Now, choose a vector $\in (\mathbf{b}_1, \ldots, \mathbf{b}_i, \mathbf{y}) \setminus (\mathbf{b}_1, \ldots, \mathbf{b}_i)$ such that

$$dist(, (\mathbf{b}_1, \ldots, \mathbf{b}_i)) \text{ is the smallest.}$$

Note, that we can do this since we have only finitely many points to choose from. Let $\mathbf{b}_{i+1} =$.

We will now show that the above algorithm returns a basis $\mathbf{b}_1, \ldots \mathbf{b}_n \in$ for the lattice. Clearly all $\mathbf{b}_i s \in$ and they are linearly independent by the algorithm that we used. We are left to show that

$$\subseteq \{\Sigma x_i \mathbf{b}_i : x_i \in \mathbb{Z}\}.$$

Let $= \Sigma z_i \mathbf{b}_i$ be an arbitrary lattice vector (where $z_i \in \mathbb{R}$). Let $= \Sigma \lfloor z_i \rfloor \mathbf{b}_i \in$. Then, $- = \Sigma (z_i - \lfloor z_i \rfloor) \mathbf{b}_i \in$. We will show that all coefficients z_i must be integers. Express

$$- = (z_n - \lfloor z_n \rfloor)\mathbf{b}_n + (\mathbf{b}_1, \ldots, \mathbf{b}_{n-1}) = (z_n - \lfloor z_n \rfloor)\tilde{\mathbf{b}_n} + (\mathbf{b}_1, \ldots, \mathbf{b}_{n-1})$$

In other words, vector $-$ is in the $(\mathbf{b}_1, \ldots, \mathbf{b}_{n-1})$ plus a multiple of $\tilde{\mathbf{b}_n}$ with coefficients $0 \leq \lfloor z_n \rfloor < 1$.
Now,

$$dist(-, (\mathbf{b}_1, \ldots, \mathbf{b}_{n-1})) = (z_n - \lfloor z_n \rfloor)||\tilde{\mathbf{b}_n}||$$

This follows because the distance is defined as the orthogonal component of $-$ to the span $(\mathbf{b}_1, \ldots, \mathbf{b}_{n-1})$, which is precisely $(z_n - \lfloor z_n \rfloor)||\tilde{\mathbf{b}_n}||$. Similarly,

$$dist(\mathbf{b}_n, (\mathbf{b}_1, \ldots, \mathbf{b}_{n-1})) = ||\tilde{\mathbf{b}_n}||$$

In addition, since $0 \leq (z_n - \lfloor z_n \rfloor) < 1$,

$$dist(-, (\mathbf{b}_1, \ldots, \mathbf{b}_{n-1})) < dist(\mathbf{b}_n, (\mathbf{b}_1, \ldots, \mathbf{b}_{n-1}))$$

But since $\mathbf{b}_n$ was chosen as the closest vector to $(\mathbf{b}_1, \ldots, \mathbf{b}_{n-1})$, $-$ must be linearly dependent of $\mathbf{b}_1, \ldots, \mathbf{b}_{n-1}$. Therefore, $z_n - \lfloor z_n \rfloor = 0$ and so $z_n \in \mathbb{Z}$. By recursively repeating the above argument for $= -z_i \mathbf{b}_i \in (\mathbf{b}_1, \ldots, \mathbf{b}_{i-1})$ for all $1 < i \leq n$ we obtain that all coefficients z_j for $1 \leq j \leq n$ must be integers. $\square$

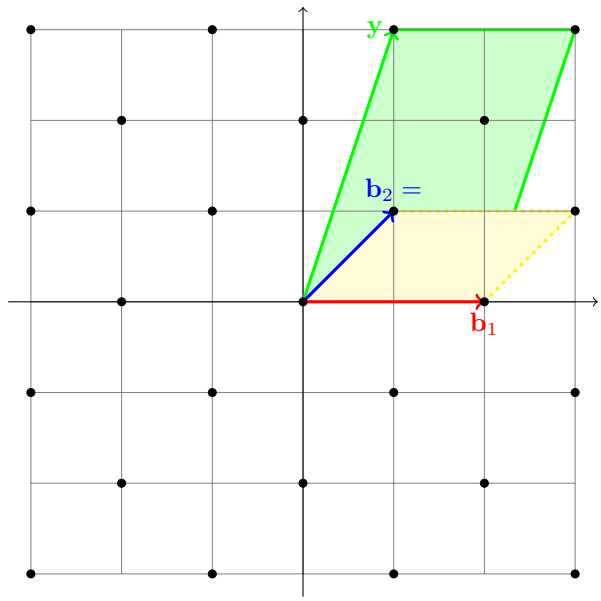


Figure 1: Constructing lattice basis from a discrete additive subgroup of . After the first iteration if we choose $\mathbf{y} = (1, 3)$, then in the $(\mathbf{b}_1, \mathbf{y})$ we choose $= \mathbf{b}_2$ which is at the minimum distance from $(\mathbf{b}_1)$. We can see that there are no non-zero lattice vectors in $(\mathbf{b}_1, \mathbf{b}_2)$. Therefore, $\{\mathbf{b}_1, \mathbf{b}_2\}$ forms a basis for this lattice.

3 Upper bound on a shortest lattice vector

Last time we saw how to compute a lower bound on the length of a shortest lattice vector. In particular, we saw that

$$\geq \min_{i=1,\ldots,n} ||\tilde{\mathbf{b}}||$$

Now, we will compute an upper bound on .

Before we state the theorem, we give some intuition on what a shortest lattice vector might depend on. Recall, that the determinant of a lattice is an inverse of its density. So larger determinant implies less dense lattice, while smaller determinant implies denser lattice. Therefore, we should be able to express an upper bound for in terms of the determinant of a lattice. But we know that $\leq det()$. We will now prove a stronger bound. In addition, our expression should scale well. In particular, if is an arbitrary lattice with a shortest vector of length , then lattice defined by scaling every vector in by k should have a shortest vector of length k. Similarly, $det(c) = c^n$ should satisfy. One can check that Minkowski's first theorem holds well on these properties.

Theorem 5 (Minkowski's First Theorem)**.** *For every full rank lattice :*

$$\leq \sqrt{n} * det()^{1/n}.$$

In order to prove the theorem, we will need to use the following 2 theorems.

Theorem 6 (Blichfeld)**.** *For all full rank lattice and measurable set $S \subseteq$ s.t. $vol(S) > det()$,*

$$\exists \mathbf{x}, \mathbf{y} \in S, s.t.\ \mathbf{x} - \mathbf{y} \in .$$

See Figure-2 for an example.

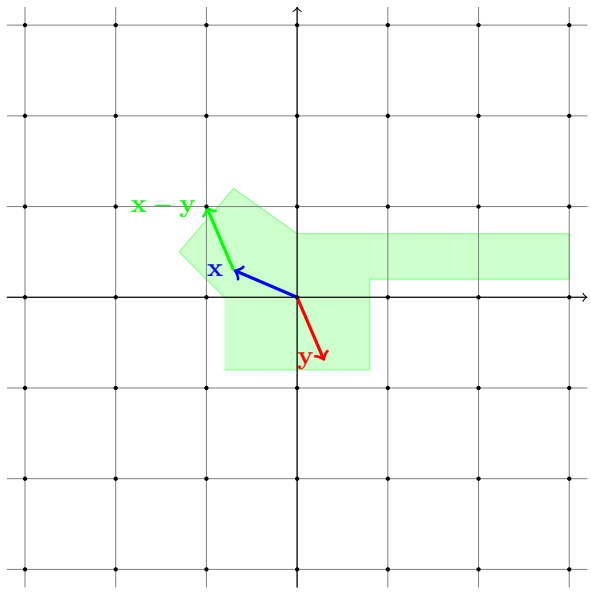


Figure 2: By the Blichfeld's theorem we can find $\mathbf{x}$ and $\mathbf{y}$ in this set such that $\mathbf{x} - \mathbf{y} \in$.

Proof. Let B be a basis for the lattice . Define $f :\to (B)$ as follows: $f(\Sigma x_i \mathbf{b}_i) = \Sigma(x_i - \lfloor x_i \rfloor)\mathbf{b}_i$. First, note that $\Sigma x_i \mathbf{b}_i - f(\Sigma x_i \mathbf{b}_i) = \Sigma \lfloor x_i \rfloor \mathbf{b}_i \in$. Now consider the following 2 cases:

Case 1: If $\exists \mathbf{x}, \mathbf{y} \in S$ s.t. $f(x) = f(y)$ (i.e. we have a collision from two vectors). Then, $\mathbf{x} - \mathbf{y} = (\mathbf{x} - f(\mathbf{x})) - (\mathbf{y} - f(\mathbf{y}))$. But as noted above, $\mathbf{x} - f(\mathbf{x}) \in$ and $\mathbf{y} - f(\mathbf{y}) \in$. Therefore, $\mathbf{x} - \mathbf{y} \in$.

Case 2: Assume there are no collisions. Let $S = \bigcup_{\mathbf{x}\in} S_x$. Define $\tilde{S_\mathbf{x}} = S_\mathbf{x} - \mathbf{x}$. By definition, $\tilde{S_\mathbf{x}} \subseteq (B)$. Also, $vol(S) = \Sigma vol(S_\mathbf{x})$ and $vol(\tilde{S_\mathbf{x}}) = vol(S_\mathbf{x})$. Therefore, $vol(S) = \Sigma vol(S_\mathbf{x}) = \Sigma vol(\tilde{S_\mathbf{x}})$. But since we assume that we do not have any collisions, then for all $\mathbf{x}, \mathbf{y}$, $\tilde{S_\mathbf{x}} \bigcap \tilde{S_\mathbf{y}} = \emptyset$. And so,

$$vol(S) = \Sigma vol(\tilde{S_\mathbf{x}}) = vol(\bigcup_{\mathbf{x}\in} \tilde{S_\mathbf{x}}) \leq vol((B)) = det()$$

Therefore, $vol(S) \leq det()$ which contradicts with our assumption. □

Definition 7 (Convex Set)**.** *A set S is convex if:*

$$\forall x \neq y \in S, \forall \alpha \in [0,1], \alpha x + (1-\alpha)y \in S.$$

Informally, the above definition say that if we take any two points from the set, any point that lies on the straight line between the two points must also be in the set.

Definition 8 (Centrally Symmetric Set)**.** *A set S is centrally symmetric if:*

$$\forall x \in S, -x \in S.$$

Theorem 9 (Minkowski's Convex Body Theorem)**.** *For all full-rank lattice , and a convex centrally symmetric set S with $vol(S) > 2^n det()$, S contains a non-zero lattice point.*

Proof. Let $\tilde{S} = \{\mathbf{x}/2 : \mathbf{x} \in S\}$. Then,

$$vol(\tilde{S}) = 2^{-n} vol(S) > det()$$

Therefore, by the Blichfeld's theorem $\exists \mathbf{x}, \mathbf{y} \in \tilde{S}$ s.t. $\mathbf{x} - \mathbf{y} \in$. We will show that $\mathbf{x} - \mathbf{y} \in S$. Now, $2\mathbf{x} \in S$ and $2\mathbf{y} \in S$ by the construction of $\tilde{S}$. Therefore, $-2\mathbf{y} \in S$. And $\mathbf{x} - \mathbf{y} = \frac{2\mathbf{x}-2\mathbf{y}}{2} \in S$.

□

We are now ready to prove Minkowski's first theorem (Theorem 1).

Proof. (Minkowski's first theorem) Let $S = \mathcal{B}(0,)$, where $\mathcal{B}(\mathbf{x}, r)$ is an n-dimensional open ball of radius r centred at $\mathbf{x}$.
Note that using l_2 norm this n-dimensional ball contains an n-dimensional cube of length $\frac{2r}{\sqrt{n}}$. Therefore,

$$vol(\mathcal{B}(0, r)) \geq (\frac{2r}{\sqrt{n}})^n.$$

Therefore, we get $vol(\mathcal{B}(0,)) \geq (\frac{2}{\sqrt{n}})^n$. But from Minkowski's convex body theorem and the fact that the ball is open and hence contains no non-zero lattice points, we get

$$(\frac{2}{\sqrt{n}})^n \leq vol(\mathcal{B}(0,) \leq 2^n det().$$

Rearranging,

$$\leq \sqrt{n} * det()^{1/n}.$$

□

4 Successive Minima

Given a lattice, it is natural to ask questions such as: What is a shortest vector in the lattice? What is a second shortest vector in the lattice? In general, what is the length of the ith shortest vector in the lattice? As we saw from the last lecture, the length of a shortest lattice vector is defined as . We now extend this definition:

Definition 10 (Successive Minima)**.** *Let be an arbitrary lattice of rank n. Then $\forall i, 1 \leq i \leq n$:*

$$\stackrel{\text{def}}{=} inf\{r : \mathcal{B}(0, r) \text{ contains } \geq i \text{ linearly independent lattice vectors}\}$$

Following the above the definition and the lattice described in Figure-3, we can see that $= ||\mathbf{x}_1||$ and $= ||\mathbf{x}_3||$, since neither $\mathbf{x}_2$ nor $2\mathbf{x}_1$ are linearly independent of $\mathbf{x}_1$.

We saw that Minkowski's first theorem gives us an upper bound on . In fact, his second theorem strengthens the results by considering a geometric mean of , , . . . ,. We leave the proof of the theorem to the reader.

Theorem 11 (Minkowski's Second Theorem)**.** *For all full rank lattices ,*

$$(\prod_{i=1}^{n})^{1/n} \leq \sqrt{n} * (det())^{1/n}.$$

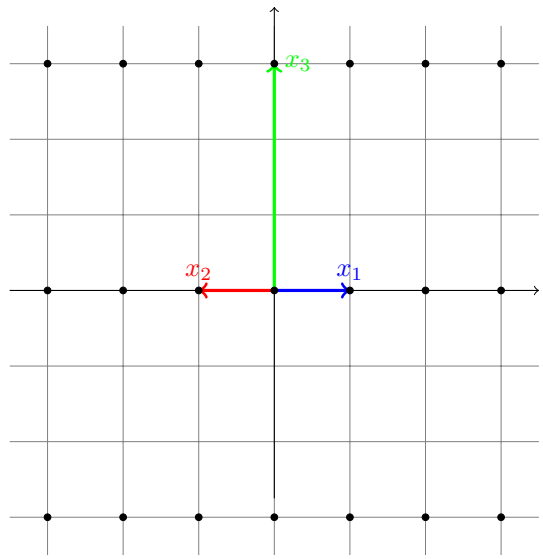


Figure 3: The first and second successive minima in the lattice generated by $(1, 0)$ and $(0, 3)$. Knowing that $= ||\mathbf{x}_1||$, we can ask whether $= ||\mathbf{x}_2||, = 2$ or $= ||\mathbf{x}_3||$. By the definition, $= ||\mathbf{x}_3||$.

5 Applications of Minkowski's Theorem

Minkowski's Theorem is widely used in computer and mathematical sciences. For example, we now can prove the following theorems:

- Dirichlets theorem on Diophantine approximation. This theorem allows us to approximate real numbers with rationals (See Figure-4 for an example).
- Lagranges four-square theorem. Intuitively, the theorem states that we can express every positive integer as the sum of four squares of integers. We leave the proof of this theorem to the reader.

Theorem 12 (Dirichlet's Theorem on Diophantine Approximation)**.** *For all $\lambda \in \mathbb{R}$ and all $Q \in \mathbb{N}$, there exists p, q such that*

$$q < Q \text{ and } |\lambda - \frac{p}{q}| \leq \frac{1}{qQ}.$$

Proof. Consider the lattice $=$. Let $S = \{(x, y) : -Q \leq x \leq Q, -\frac{1}{Q} \leq \lambda x - y \leq \frac{1}{Q}\}$.
Now, $vol(S) =$ base*height$= \frac{2}{Q} * 2Q = 4$. Therefore, $vol(S) \geq 2^2 det()$.
Applying Minkowski's first theorem, we know that there exists $(q, p) \in$ such that,

$$-Q \leq q \leq Q \text{ and } -\frac{1}{Q} \leq \lambda q - p \leq \frac{1}{Q}.$$

Therefore, $|\lambda q - p| \leq \frac{1}{Q}$, which implies $|\lambda - \frac{p}{q}| \leq \frac{1}{qQ}$. $\square$

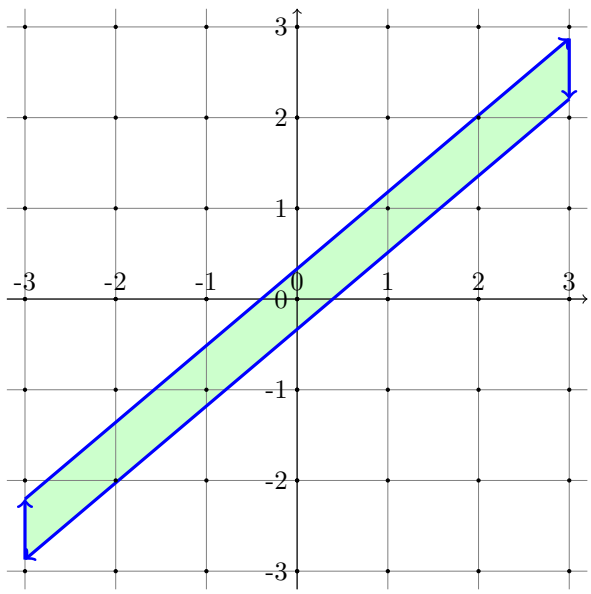


Figure 4: Applying Dirichlet's theorem to approximate 0.846153846 for $Q = 3$, we can see that there exists (p, q) satisfying the requirements. For example, $(p, q) = (1, 1)$, since $1 \leq 3$ and $|0.846153846 - 1| = 0.153846154 \leq \frac{1}{1*3}$.

6.876 Advanced Topics in Cryptography: Lattices	September 16, 2015

Lecture 3

Lecturer: Vinod Vaikuntanathan — *Scribe: Itay Berman*

In the first two lectures we introduces lattices and covered the following topics:

- Minkowski's first and second convex body theorems.
- Gram-Schmidt Orthogonalization and used it to lower-bound the length of the shortest non-zero vector in a lattice ($\lambda_1 \geq \min_i \|\mathbf{b}_i\|$).
- Proof for the four squares theorem using lattices.

In the next few lectures we will cover:

- Computational problem with lattices (SVP, CVP, SIVP).
- Overview of the complexity landscape.
- The LLL algorithm, which finds an approximation to the shortest vector. We will then use the LLL algorithm to find the shortest vector exactly (while increasing the running time), and to find an approximation to the closest vector problem.

1 Computational Problems

We will focus on three computational problems regarding lattices:

- The Shortest Vector Problem (SVP): find the shortest non-zero vector in the lattice.
- The closest Vector Problem (CVP): find the closest lattice vector to a given vector.
- The Shortest Independent Vectors Problem (SIVP): find n independent and "short" vectors.

We define search, approximation and decision variants of the above problems. Recall that $\lambda_1 = \lambda_1(\mathbf{B})$ is the first successive minima of $\mathcal{L}(\mathbf{B})$ (i.e., is the length of the shortest non-zero vector in $\mathcal{L}(\mathbf{B})$), that λ_n is the n'th successive minima of $\mathcal{L}(\mathbf{B})$ and that $\mathrm{dist}(t, \mathcal{L}(\mathbf{B})) = \min_{u \in \mathcal{L}(\mathbf{B})} \|u - t\|$.

Search	Approximation for $\gamma > 1$	Decision (Promise[1]) for $\gamma > 1$
SVP: given a basis $\mathbf{B} \in \mathbb{Z}^{n \times n}$, find $u \in \mathcal{L}(\mathbf{B}) \setminus \{0\}$ s.t. $\|u\| = \lambda_1$.	**SVP$_\gamma$:** given a basis $\mathbf{B} \in \mathbb{Z}^{n \times n}$, find $u \in \mathcal{L}(\mathbf{B}) \setminus \{0\}$ s.t. $\|u\| \leq \gamma \cdot \lambda_1$.	**gapSVP$_\gamma$:** $YES = \{(\mathbf{B}, s) \colon \lambda_1(\mathbf{B}) \leq s\}$, $NO = \{(\mathbf{B}, s) \colon \lambda_1(\mathbf{B}) > \gamma \cdot s\}$
CVP: given a basis $\mathbf{B} \in \mathbb{Z}^{n \times n}$ and $t \in \mathbb{Z}^n$, find $u \in \mathcal{L}(\mathbf{B})$ s.t. $\|u - t\| = \mathrm{dist}(\mathcal{L}(\mathbf{B}), t)$.	**CVP$_\gamma$:** given a basis $\mathbf{B} \in \mathbb{Z}^{n \times n}$ and $t \in \mathbb{Z}^n$, find $u \in \mathcal{L}(\mathbf{B})$ s.t. $\|u - t\| \leq \gamma \cdot \mathrm{dist}(\mathcal{L}(\mathbf{B}), t)$.	**gapCVP$_\gamma$:** $YES = \{(\mathbf{B}, t, s) \colon \mathrm{dist}(\mathcal{L}(\mathbf{B}), t) \leq s\}$, $NO = \{(\mathbf{B}, t, s) \colon \mathrm{dist}(\mathcal{L}(\mathbf{B}), t) > \gamma \cdot s\}$
SIVP: given a basis $\mathbf{B} \in \mathbb{Z}^{n \times n}$, find independent vectors $u_1, \ldots, u_n$ s.t. $\|u_i\| \leq \lambda_n$ for $i \in [n]$.	N/A	N/A

[1] A promise problem is such that the input domain is $YES \cup NO$. An algorithm decides a promise problem if it accepts inputs from YES and rejects inputs from NO.

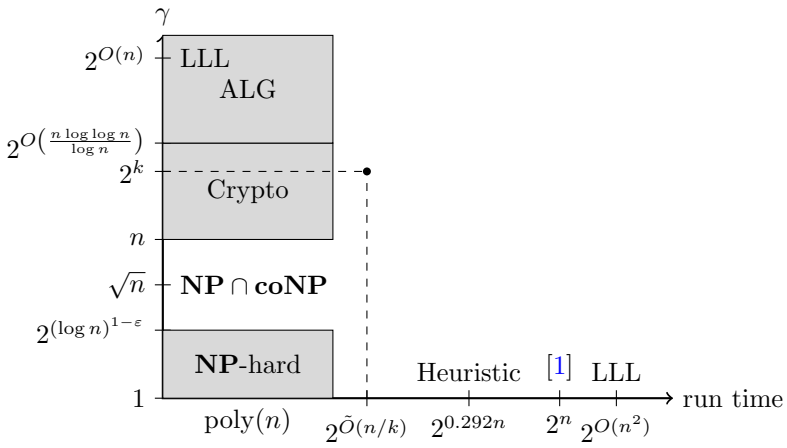


Figure 1: The complexity landscape of SVP_γ.

2 Complexity Landscape

We present the landscape for the Shortest Vector Problem. The landscape for the Closest Vector Problem is very similar. In the following we assume we are given a lattice basis in $\mathbb{Z}^{n\times n}$. The running times of the described algorithms will be a function of n, and we ignore polynomial factors in the length of the bit representation of the given basis.

Algorithms for SVP_γ: The first algorithm to solve the SVP_γ was the LLL algorithm [2]. It gave a $2^{O(n)}$-approximation and its running time is poly(n). The best know approximation factor achieved by a polynomial time algorithm is $2^{O(\frac{n \log\log n}{\log n})}$.

If we want to solve the exact SVP (i.e., $\gamma = 1$), then the LLL algorithm can do so with running time of $2^{O(n^2)}$. The fastest known algorithm to solve the exact SVP was recently given in [1], and its running time is 2^n. There are also Heuristics algorithms with running time $2^{0.292n}$.

Finally, One can have a trade off between the approximation factor and the running time — achieving 2^k-approximation can be done in $2^{\tilde{O}(n/k)}$ time.

Hardness of SVP_γ: It is no surprise that we don't know of a polynomial time algorithm to solve the exact SVP, since it was shown to be **NP**-hard. In fact, even achieving $2^{(\log n)^{1-\varepsilon}}$-approximation is **NP**-hard. On the other hand, it was shown that $\mathsf{SVP}_{\sqrt{n}}$ is in $\mathbf{NP} \cap \mathbf{coNP}$, and thus unlikely to be **NP**-hard.

Cryptography from SVP_γ: The smallest approximation factor from which we know how to build cryptographic primitives is $\gamma = n$. Since $\mathsf{SVP}_{\sqrt{n}} \in \mathbf{NP} \cap \mathbf{coNP}$, this will likely not suffice to base cryptography on **NP**-hardness.

A pictorial presentation of the above description if given in Figuire 1. Throughout this course we'll cover all the aforementioned results, starting with the LLL algorithm. However, before presenting the LLL algorithm, we warm up by reducing the Shortest Vector Problem to the Closest Vector Problem.

Theorem 1. *For any $\gamma > 1$, given access to CVP_γ oracle, there exists a polynomial time algorithm that solves SVP_γ.*

Proof. We prove for the case $\gamma = 1$, and the proof generalizes to $\gamma > 1$ naturally. For a basis $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_n]$ and $i \in [n]$, let $\mathbf{B}_i = [\mathbf{b}_1, \ldots, \mathbf{b}_{i-1}, 2\mathbf{b}_i, \mathbf{b}_{i+1}, \ldots, \mathbf{b}_n]$. The polynomial time algorithm that solves SVP sets $v_i := \mathsf{CVP}(\mathbf{B}_i, \mathbf{b}_i)$ for every $i \in [n]$, and return $v_{i^*} - \mathbf{b}_{i^*}$ such that $i^* = \arg\min_i \|v_i - \mathbf{b}_i\|$.

Let $v = \sum_{i=1}^n c_i \mathbf{b}_i$ be the shortest vector in $\mathcal{L}(\mathbf{B})$. We show that the algorithm returns v. The correctness of the algorithm follows from the next claim.

Claim 2. *Assume that c_j is odd, then $v + \mathbf{b}_j$ is the closest vector to $\mathbf{b}_j$ in $\mathcal{L}(\mathbf{B}_j)$.*

Proof. First, it is easy to see that $v + \mathbf{b}_j \in \mathcal{L}(\mathbf{B}_j)$. Assume towards a contradiction that there exists $v' \in \mathcal{L}(\mathbf{B}_j)$ such that $\|v' - \mathbf{b}_j\| < \|(v + \mathbf{b}_j) - \mathbf{b}_j\| = \|v\|$. But, it is easy to see that $v' - \mathbf{b}_j$ belongs to $\mathcal{L}(\mathbf{B})$, a contradiction to the fact that v is the shortest vector in $\mathcal{L}(\mathbf{B})$. □

Since at least one of the c_i's is odd (as otherwise $v/2$ would be a shorter lattice vector), say c_j, the algorithm will set $v_j = v + \mathbf{b}_j$, and will return v. □

3 The LLL Algorithm

We begin describing the algorithm of Lenstra, Lenstra and Lovász for approximating the shortest vector of a lattice.

Theorem 3. *Given a basis $\mathbf{B} \in \mathbb{Z}^{n \times n}$ there is a $\mathrm{poly}(n, W_{\mathbf{B}})$-time algorithm for $\mathsf{SVP}_{2^{O(n)}}$, where $W_{\mathbf{B}}$ is the length of the bit representation of $\mathbf{B}$; namely, the algorithm returns a vector $v \in \mathcal{L}(\mathbf{B})$ such that $\|v\| \leq 2^{O(n)} \cdot \lambda_1(\mathbf{B})$.*

The LLL algorithm actually transforms, in polynomial time, the given basis into a "LLL-reduced" basis for the same lattice. The above theorem holds since a LLL-reduced basis has an important property — its shortest vector is a $2^{O(n)}$-approximation for the shortest vector in entire the lattice. In this lecture we'll give define a LLL-reduced basis and give some intuition for this definition; in the next lecture we'll describe and analyze the LLL algorithm itself.

3.1 LLL-reduced Basis

Our goal is to transform the given basis to one with "short" vectors. Consider the case $n = 2$, i.e., $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2]$. Our starting point is the Gram-Schmidt orthogonalization process in which we set $\tilde{\mathbf{b}}_1 := \mathbf{b}_1$ and $\tilde{\mathbf{b}}_2 := \mathbf{b}_2 - \mu_{21}\tilde{\mathbf{b}}_1$, where $\mu_{21} = \langle \mathbf{b}_2, \tilde{\mathbf{b}}_1 \rangle / \langle \tilde{\mathbf{b}}_1, \tilde{\mathbf{b}}_1 \rangle$. Intuitively, $\tilde{\mathbf{b}}_2$ is the shortest vector we can hope for, since we removed all $\tilde{\mathbf{b}}_1$'s components from it (this fact is what make the Gram-Schmidt orthogonal). However, $\tilde{\mathbf{b}}_2 \notin \mathcal{L}(\mathbf{B})$, and thus cannot be in a basis of $\mathcal{L}(\mathbf{B})$. To fix this issue, we transform $\mathbf{B}$ into the following basis: $\mathbf{b}'_1 = \mathbf{b}$ and $\mathbf{b}'_2 = \mathbf{b}_2 - \lfloor \mu_{21} \rceil \mathbf{b}'_1$, where $\lfloor \cdot \rceil$ means rounding to the closest integer. $\mathbf{b}'_2$ is the shortest *lattice* vector we can hope for, as we removed all the *integer* components of $\tilde{\mathbf{b}}_1$. Note that when projecting $\mathbf{b}'_2$ to the line generated by $\mathbf{b}'_1$, then this projection is between $-\mathbf{b}'_1/2$ to $\mathbf{b}'_1/2$. The latter fact also guarantees that the resulting basis is "close" to orthogonal — the angle between $\mathbf{b}'_1$ to $\mathbf{b}'_2$ is at least 60 degrees. See Figure 2 for an example of this transformation.

So far we reduced $\mathbf{b}_2$, but left $\mathbf{b}_1$ as is. But what if $\mathbf{b}_1$ is very long to begin with? there is no guarantee that the reduced basis is short. At this point we adopt an idea from Euclid's greatest common divisor (gcd) algorithm: we reduce $\mathbf{b}_2$ with respect to $\mathbf{b}_1$ as much as we can, then swap the roles of $\mathbf{b}_1$ and $\mathbf{b}_2$ and repeat the process. The process will stop when the basis meets the following conditions, which are the definition of LLL-reduced basis in two dimensions.

Definition 4 (LLL-reduced basis in two dimensions)**.** *A basis $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2]$ is LLL-reduced if*

1. $|\mu_{21}| \leq 1/2$.
2. $\|\mathbf{b}_2\| \geq \|\mathbf{b}_1\|$.

Note that the second condition can be written as $\left\|\tilde{\mathbf{b}}_2\right\|^2 \geq (1 - \mu_{21}^2) \left\|\tilde{\mathbf{b}}_1\right\|^2$. Generalizing for n dimensions we get the following definition.

Definition 5 (LLL-reduced basis)**.** *Let $\delta \in [1/4, 1]$. A basis $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_n]$ is δ-LLL-reduced if*

1. $|\mu_{ij}| \leq 1/2$ *for every* $1 \leq i < j \leq n$.

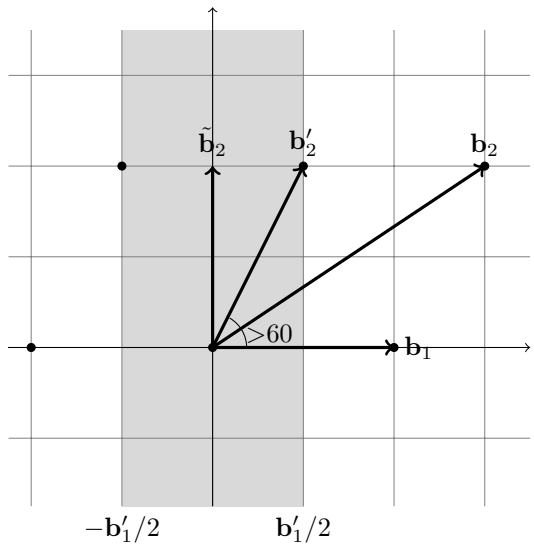


Figure 2: The LLL-reduced basis of $[\mathbf{b}_1 = (2,0), \mathbf{b}_2 = (3,2)]$ is $[\mathbf{b}'_1 = (2,0), \mathbf{b}'_2 = (1,2)]$.

2. $\left\|\tilde{\mathbf{b}}_{i+1}\right\|^2 \geq (\delta - \mu_{i+i,i}^2)\left\|\tilde{\mathbf{b}}_i\right\|^2$ *for every* $1 \leq i \leq n-1$.

Note that the projection of a (partial) LLL-reduced basis $[\mathbf{b}_1, \ldots, \mathbf{b}_{i-1}, \mathbf{b}_i, \mathbf{b}_{i+1}]$ to $\mathsf{Span}(\tilde{\mathbf{b}}_i, \ldots, \tilde{\mathbf{b}}_n)$ is $[0, \ldots, 0, \tilde{\mathbf{b}}_i, \mathbf{b}_{i+1} + \mu_{i+1,i}\tilde{\mathbf{b}}_i]$. The last two vectors meet the definition of LLL-reduced basis in two dimensions.

References

[1] Divesh Aggarwal and Daniel Dadush and Oded Regev and Noah Stephens-Davidowitz. *Solving the Shortest Vector Problem in 2^n Time Using Discrete Gaussian Sampling: Extended Abstract*. STOC 2015.

[2] Lenstra, A.K. and Lenstra, H.W.jun. and Lászlo Lovász. *Factoring polynomials with rational coefficients*. Math. Ann. 1982

6.876 Advanced Topics in Cryptography: Lattices	September 16, 2015

Lecture 4

Lecturer: Vinod Vaikuntanathan *Scribe: Akshay Degwekar*

In this class, we will cover -

- The LLL Algorithm.
- Applications of LLL -
  - Babai's Algorithm for approximating CVP.
  - Exact SVP in $2^{O(n^2)}\text{poly}(W)$ time.
  - Solving "low density" subset sum problems.

1 The LLL Algorithm

We begin describing the algorithm of Lenstra, Lenstra and Lovász for approximating the shortest vector of a lattice.

Theorem 1. *Given a basis $\mathbf{B} \in \mathbb{Z}^{n\times n}$ there is a $\text{poly}(n, W_{\mathbf{B}})$-time algorithm for $\mathsf{SVP}_{2^{O(n)}}$, where $W_{\mathbf{B}}$ is the length of the bit representation of $\mathbf{B}$; namely, the algorithm returns a vector $v \in \mathcal{L}(\mathbf{B})$ such that $\|v\| \le 2^{O(n)} \cdot \lambda_1(\mathbf{B})$.*

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1.1 LLL-reduced Basis

Our goal is to transform the given basis to one with "short" vectors. Consider the case $n = 2$, i.e., $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2]$. Our starting point is the Gram-Schmidt orthogonalization process in which we set $\tilde{\mathbf{b}}_1 := \mathbf{b}_1$ and $\tilde{\mathbf{b}}_2 := \mathbf{b}_2 - \mu_{21}\tilde{\mathbf{b}}_1$, where $\mu_{21} = \langle \mathbf{b}_2, \tilde{\mathbf{b}}_1\rangle / \langle \tilde{\mathbf{b}}_1, \tilde{\mathbf{b}}_1\rangle$. Intuitively, $\tilde{\mathbf{b}}_2$ is the shortest vector we can hope for, since we removed all $\tilde{\mathbf{b}}_1$'s components from it (this fact is what make the Gram-Schmidt orthogonal). However, $\tilde{\mathbf{b}}_2 \notin \mathcal{L}(\mathbf{B})$, and thus cannot be in a basis of $\mathcal{L}(\mathbf{B})$. To fix this issue, we transform $\mathbf{B}$ into the following basis: $\mathbf{b}'_1 = \mathbf{b}$ and $\mathbf{b}'_2 = \mathbf{b}_2 - \lfloor \mu_{21} \rceil \mathbf{b}'_1$, where $\lfloor \cdot \rceil$ means rounding to the closest integer. $\mathbf{b}'_2$ is the shortest *lattice* vector we can hope for, as we removed all the *integer* components of $\tilde{\mathbf{b}}_1$. Note that when projecting $\mathbf{b}'_2$ to the line generated by $\mathbf{b}'_1$, then this projection is between $-\mathbf{b}'_1/2$ to $\mathbf{b}'_1/2$. The latter fact also guarantees that the resulting basis is "close" to orthogonal — the angle between $\mathbf{b}'_1$ to $\mathbf{b}'_2$ is at least 60 degrees. See Figure 1 for an example of this transformation.

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1. $|\mu_{21}| \le 1/2$.

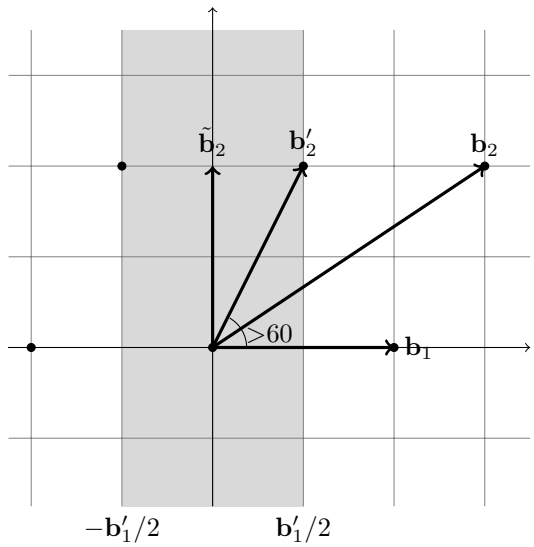

Figure 1: The LLL-reduced basis of $[\mathbf{b}_1 = (2,0), \mathbf{b}_2 = (3,2)]$ is $[\mathbf{b}'_1 = (2,0), \mathbf{b}'_2 = (1,2)]$.

2. $\|\mathbf{b}_2\| \geq \|\mathbf{b}_1\|$.

Note that the second condition can be written as $\left\|\tilde{\mathbf{b}}_2\right\|^2 \geq (1-\mu_{21}^2)\left\|\tilde{\mathbf{b}}_1\right\|^2$. Generalizing for n dimensions we get the following definition.

Definition 3 (LLL-reduced basis)**.** *Let $\delta \in [1/4, 1]$. A basis $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_n]$ is δ-LLL-reduced if*

1. $|\mu_{ij}| \leq 1/2$ *for every* $1 \leq i < j \leq n$.
2. $\left\|\tilde{\mathbf{b}}_{i+1}\right\|^2 \geq (\delta - \mu_{i+i,i}^2)\left\|\tilde{\mathbf{b}}_i\right\|^2$ *for every* $1 \leq i \leq n-1$.

Note that the projection of a (partial) LLL-reduced basis $[\mathbf{b}_1, \ldots, \mathbf{b}_{i-1}, \mathbf{b}_i, \mathbf{b}_{i+1}]$ to $\mathsf{Span}(\tilde{\mathbf{b}}_i, \ldots, \tilde{\mathbf{b}}_n)$ is $[0, \ldots, 0, \tilde{\mathbf{b}}_i, \mathbf{b}_{i+1} + \mu_{i+1,i}\tilde{\mathbf{b}}_i]$. The last two vectors meet the definition of LLL-reduced basis in two dimensions.

2 The LLL Algorithm

2.1 LLL-reduced basis

To recap, we defined a δ-LLL reduced basis last class

Definition 4 (δ-LLL-reduced basis)**.** *Let $\delta \in (1/4, 1)$. A basis $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_n]$ is δ-LLL-reduced if*

1. ***size-reduced.*** $|\mu_{i,j}| \leq 1/2$ *for every* $i > j$.
2. ***Lovász criterion.*** $\left\|\tilde{\mathbf{b}}_{i+1}\right\|^2 \geq (\delta - \mu_{i+i,i}^2)\left\|\tilde{\mathbf{b}}_i\right\|^2$ *for every* $1 \leq i \leq n-1$.

To get some intuition for this definition, we look at the 2d variant we saw last class where we said that $\|\mathbf{b}_2\| \geq \|\mathbf{b}_1\|$ equivalently $\left\|\tilde{\mathbf{b}}_2\right\|^2 \geq (1-\mu_{2,1}^2)\left\|\tilde{\mathbf{b}}_1\right\|^2$. We are slightly changing this by adding a δ as a parameter. Geometrically, the criterion looks at the projection of vectors $\mathbf{b}_1, \mathbf{b}_2 \cdots \mathbf{b}_n$ on to the Gram-Schmidt vectors. The first few vectors $\mathbf{b}_1 \cdots \mathbf{b}_{i-1}$ project to 0, $\mathbf{b}_i$ becomes $\tilde{\mathbf{b}}_i$ and $\mathbf{b}_{i+1}$ projects to

$\tilde{\mathbf{b}}_{i+1} + \mu_{i+1,i}\tilde{\mathbf{b}}_i$. So the Lovász criterion compares the norms of these two projected vectors and says that the second one is 'longer' than the first one, like the 2d case.

It is interesting to note that this condition is extremely local and it can be extended to looking at k vectors at a time to give a 2^k time, $2^{n/k}$ approximation to SVP. We want to argue that finding a LLL-reduced basis is enough to get an approximate shortest vector.

Lemma 5. *If B is a δ-LLL reduced basis, then* $\|b_1\| \leq \left(\frac{2}{\sqrt{4\delta-1}}\right)^{n-1} \lambda_1$

Proof. Since B is δ-LLL reduced, we know that -

$$\left\|\tilde{\mathbf{b}}_{i+1} + \mu_{i+1,i}\tilde{\mathbf{b}}_i\right\|^2 \geq \delta \left\|\tilde{\mathbf{b}}_i\right\|$$

Using the fact that the Gram-Schmidt basis is ortogonal, we rearrange the equation to get

$$\left\|\tilde{\mathbf{b}}_{i+1}\right\|^2 \geq (\delta - \mu_{i+1,i}^2) \left\|\tilde{\mathbf{b}}_i\right\|^2$$

Here we substitute the fact that $|\mu_{i+1,i}| \leq \frac{1}{2}$ to get that

$$\left\|\tilde{\mathbf{b}}_{i+1}\right\|^2 \geq (\frac{4\delta - 1}{4}) \left\|\tilde{\mathbf{b}}_i\right\|^2$$

From this we infer that $\left\|\tilde{\mathbf{b}}_i\right\|^2 \geq (\frac{4\delta-1}{4})^{i-1} \left\|\tilde{\mathbf{b}}_1\right\|^2$. This used in conjunction with $\lambda_1 \geq \min_i \left\{\left\|\tilde{\mathbf{b}}_i\right\|\right\}$ gives us the required result. □

So, in any LLL-reduced basis, the first vector would be a good approximation to the shortest vector. It is interesting to note that this gives us a bit more - we can use the fact that the i-th largest element of the set - $\left\{\left\|\tilde{\mathbf{b}}_j\right\|\right\}_j$ gives us a lower bound on λ_i to get a comparable approximation on λ_i for all i.

2.2 Finding an LLL-reduced basis

Lemma 5 reduces the problem of getting a $2^{O(n)}$ approximation to SVP to finding an LLL-reduced basis. In this section we describe the LLL algorithm to find a reduced basis and analyze it.

Input Basis $\mathbf{b}_1, \mathbf{b}_2 \cdots \mathbf{b}_n$
while
1 Compute $\tilde{\mathbf{b}}_1, \tilde{\mathbf{b}}_2 \cdots \tilde{\mathbf{b}}_n$ [a]
2 **for** $i = 2$ to n // Reduction Step
for $j = i - 1$ to 1
$\mathbf{b}_i \leftarrow \mathbf{b}_i - c_{i,j}\mathbf{b}_j$ where $c_{i,j} = \lfloor \mu_{i,j} \rceil$
3 **if** $\exists$ i such that $\left\|\tilde{\mathbf{b}}_{i+1}\right\| < \sqrt{\delta - \mu_{i+1,i}^2} \left\|\tilde{\mathbf{b}}_i\right\|$ [b] // Swap step
Swap $\mathbf{b}_i$ and $\mathbf{b}_{i+1}$
else
Output $\mathbf{b}_1, \mathbf{b}_2 \cdots \mathbf{b}_n$

[a] This step does have some issues with runtime - the exact Gram-Schmidt vectors might keep getting bigger in terms of bit length. We don't consider those issues here.

[b] We do not recompute the $\tilde{\mathbf{b}}_i$'s again because the Reduction step does not affect $\tilde{\mathbf{b}}_i$'s.

Figure 2: The LLL algorithm

The LLL-algorithm is an iterative algorithm where in each iteration, we first replace the vectors $\mathbf{b}_i$'s by 'approximately' orthogonal vectors in the lattice obtained from the vectors $\tilde{\mathbf{b}}_i$'s. After this, we check if the Lovász criterion is violated for any pair of vectors $\mathbf{b}_i, \mathbf{b}_{i+1}$. In case of a violation, we swap the two vectors and continue.

Correctness To prove correctness, we need to show that, if the algorithm terminates, then the output basis is LLL-reduced.

In order to show that $|\mu_{i,j}| \leq \frac{1}{2}$, consider the last iteration. It will not do any swaps and hence the vectors $\tilde{\mathbf{b}}_i$'s will remain unchanged in the iteration. On the other hand, because we subtract $c_{i,j}$'s, the values of $\mu_{i,j}$'s changes. Namely $\mu_{i,j}^{new} = \mu_{i,j}^{old} - \lfloor \mu_{i,j}^{old} \rceil$ and hence $\left|\mu_{i,j}^{new}\right| \leq 1/2$. Note that this relies on the fact that we are decrementing j in step 2 of the algorithm. The Lovász criterion is satisfied by termination – if it was not satisfied for some i, then we would swap $\mathbf{b}_i$ and $\mathbf{b}_{i+1}$ and iterate again.

Termination The termination is a potential argument. We define a non-negative potential function $\phi(\mathbf{B})$ for any basis and then show that it was not too large to begin with and that each iteration reduces this function by a constant.

Let $\phi(\mathbf{B}) = \prod_i \phi_i(\mathbf{B})$ where

$$\phi_i(\mathbf{B}) = |\det(\mathcal{L}(\mathbf{b}_1, \mathbf{b}_2 \cdots \mathbf{b}_i))|$$

Where the determinant for non full rank matrices is defined as $\det(\mathbf{B}) = \det(\sqrt{\mathbf{B}^\top \mathbf{B}})$. Another way to write it would be

$$\phi_i(\mathbf{B}) = \left\|\tilde{\mathbf{b}}_1\right\| \cdot \left\|\tilde{\mathbf{b}}_2\right\| \cdots \left\|\tilde{\mathbf{b}}_i\right\|$$

Observation 6. *$\phi(\mathbf{B})$ is not too large to begin with*

$$\phi(\mathbf{B}_{init}) \leq \prod_{i=1}^{n} \|\mathbf{b}_1\| \cdot \|\mathbf{b}_2\| \cdots \|\mathbf{b}_i\| \leq \max_i (\|\mathbf{b}_i\|)^{O(n^2)}$$

So, $\log(\phi(\mathbf{B}_{init})) = \text{poly}(n, W)$ where W is the bit length of the vectors. We also know that $\phi(\mathbf{B}) \geq 1$ because of the fact that we are dealing with integer lattices and each potential ϕ_i can be interpreted as the i-dimensional volume enclosed by the vectors $\mathbf{b}_1, \mathbf{b}_2 \cdots \mathbf{b}_i$.

Observation 7. *The reduction step does not change the potential function*

This is evident by looking at $\phi_i(\mathbf{B}) = \left\|\tilde{\mathbf{b}}_1\right\| \left\|\tilde{\mathbf{b}}_2\right\| \cdots \left\|\tilde{\mathbf{b}}_i\right\|$ and observing that the reduction step leaves $\tilde{\mathbf{b}}_i$'s invariant.

Observation 8. *The swap step reduces ϕ by a constant factor.*

Proof. Let us say that $\mathbf{b}_i$ and $\mathbf{b}_{i+1}$ were swapped. This only affects the value of $\phi_i(\mathbf{B})$ because changing order of vectors does not affect the determinant.

The old value of ϕ_i is, $\phi_i^{old} = \left\|\tilde{\mathbf{b}}_1\right\| \cdot \left\|\tilde{\mathbf{b}}_2\right\| \cdots \left\|\tilde{\mathbf{b}}_i\right\|$ while the new value is $\phi_i^{new} = \left\|\tilde{\mathbf{b}}_1\right\| \cdot \left\|\tilde{\mathbf{b}}_2\right\| \cdots \left\|\tilde{\mathbf{b}}_{i-1}\right\| \cdot \left\|\tilde{\mathbf{b}}_{i+1}\right\|$. We see that the component of $\tilde{\mathbf{b}}_{i+1}$ orthogonal to the span of $\mathbf{b}_1, \mathbf{b}_2 \cdots \mathbf{b}_{i-1}$ is $\tilde{\mathbf{b}}_{i+1} + \mu_{i+1,i}\tilde{\mathbf{b}}_i$. So,

$$\frac{\phi^{new}}{\phi^{old}} = \frac{\phi_i^{new}}{\phi_i^{old}} = \frac{\left\|\tilde{\mathbf{b}}_{i+1} + \mu_{i+1,i}\tilde{\mathbf{b}}_i\right\|}{\left\|\tilde{\mathbf{b}}_i\right\|} < \sqrt{\delta}$$

So as long as $\delta < 1$ is a fixed constant, we know that the potential function decreases by a constant factor. □

Combining the observations we see that the algorithm has to terminate in time $\text{poly}(n, W)$.

3 Applications

The LLL algorithm has many applications - the most famous one being factoring polynomials over rationals. It also gives us an algorithm for finding the exact shortest vector - while the algorithm takes exponential time, it was the first algorithm to solve exact SVP for a fixed dimension.

3.1 Computing Shortest Vectors in $2^{O(n^2)}$ Time

Corollary 9 (Exact SVP)**.** *There is an exact SVP algorithm running in time* $2^{O(n^2)}$

Proof. To show this we consider an LLL-reduced basis $\mathbf{B}$. Consider a shortest vector $v = \sum_i c_i \mathbf{b}_i$. We want to say that $c_i \leq 2^{O(n)}$ and hence by iterating over c_i's and returning the smallest vector would give us a $2^{O(n^2)}$ algorithm.

Consider $v = \sum_i c_i \mathbf{b}_i = v = \sum_i \tilde{c}_i \tilde{\mathbf{b}}_i$. We look at the last i such that c_i is non-zero. Then $c_i = \tilde{c}_i$ because it is the only coefficient contributing in the $\tilde{\mathbf{b}}_i$ direction. Since $\|\mathbf{b}_1\| \geq \|v\| \geq c_i \left\|\tilde{\mathbf{b}}_i\right\|$, we get that $c_i \leq (\frac{4}{4\delta-1})^{(i-1)/2}$ because $\left\|\tilde{\mathbf{b}}_i\right\|^2 \geq (\frac{4\delta-1}{4})^{i-1} \left\|\tilde{\mathbf{b}}_1\right\|^2$. □

3.2 Babai's Algorithms for Approximate CVP

Corollary 10 (Babai's Algorithm for Approximate CVP)**.** *Let* $\mathbf{B}$ *be a reduced LLL-basis and* $\boldsymbol{y}$ *be the input vector. There are two variants of the algorithm -*

1. *Babai's rounding algorithm - We find the representation of* $\boldsymbol{y}$ *in the real span of* $\mathbf{B}$ *and output the rounded coefficients. Succinctly, output* $\mathbf{B} \lfloor \mathbf{B}^{-1} y \rceil$
2. *Babai's nearest plane algorithm - The algorithm works iteratively. We consider the hyperplane* $H = \mathsf{Span}(\mathbf{b}_1 \cdots \mathbf{b}_{n-1})$ *and its translations -* $\mathbf{b}_n + H$, $2\mathbf{b}_n + H$ *and so on and find the one closest to* $\boldsymbol{y}$*, then project* $\boldsymbol{y}$ *on* H *and recurse.*

References

[1] Divesh Aggarwal and Daniel Dadush and Oded Regev and Noah Stephens-Davidowitz. *Solving the Shortest Vector Problem in 2^n Time Using Discrete Gaussian Sampling: Extended Abstract.* STOC 2015.

[2] Lenstra, A.K. and Lenstra, H.W.jun. and Lászlo Lovász. *Factoring polynomials with rational coefficients.* Math. Ann. 1982